

Overview of Logistic Regression

- **Simplest Logistic Regression**

Learning Objectives

- Apply the general theory of the maximum likelihood estimator (MLE) to the simplest logistic regression model
- Construct the confidence interval of model parameters

Simplest Logistic Regression

- **Data**: $(Y_i, x_{i1}, \dots, x_{i,p-1})$ for $i = 1, \dots, n$, where $Y_i \in \{0, 1\}$ is the class label.
- **Logistic Regression Model** is defined as two components:
 - $P(Y_i = 1) = \pi_i$ and $P(Y_i = 0) = 1 - \pi_i$
 - $\log \frac{\pi_i}{1 - \pi_i} = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_{p-1} x_{i,p-1}$
- **Simplest logistic Regression Model**: when $p = 2$ and $x_{i1} \in \{0, 1\}$
 - The explicit solution of the MLE exists
 - It helps us to better understand the general theory

Data in Simplest Logistic Regression

Data: (Y_i, x_i) for $i = 1, \dots, n$

where $Y_i \in \{0, 1\}$, $x_i \in \{0, 1\}$

Example:

	Y	X
[1,]	0	0
[2,]	0	0
[3,]	1	0
[4,]	0	1
[5,]	0	0
[6,]	1	1

.....

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**The data can be summarized
as 2x2 table (this is typical
in biostatistics)**

$$n = a + b + c + d$$

	$Y_i = 1$	$Y_i = 0$	Sum
$x_i = 1$	a	b	$a + b$
$x_i = 0$	c	d	$c + d$

Is X and Y associated?

	$Y = 1$	$Y = 0$	Sum
$X = 1$	a	b	$a + b$
$X = 0$	c	d	$c + d$

- **Data**: (Y_i, x_i) for $i = 1, \dots, n$ where $Y_i \in \{0, 1\}$, $x_i \in \{0, 1\}$.
In biostatistics, X refers as the exposure (e.g., does the subject smoke or not), and Y refers as the disease (e.g., lung cancer).
- **Question**: based on this 2×2 table, is X associated with Y ?

Simplest Logistic Regression

- The Simplest Logistic Regression Model:

- $P(Y_i = 1) = \pi_i, \quad P(Y_i = 0) = 1 - \pi_i$

- $\log \frac{\pi_i}{1-\pi_i} = \beta_0 + \beta_1 x_i$

- Testing whether X is associated with Y is equivalent to testing

$$H_0: \beta_1 = 0 \text{ v.s. } H_1: \beta_1 \neq 0$$

- If we can find a 95% confidence interval (CI) of β_1 , say,

$$\hat{\beta}_1 \pm 1.96 \sqrt{\widehat{\text{Var}}(\hat{\beta}_1)}$$

then we reject $H_0: \beta_1 = 0$ if and only if 0 is outside this CI.

The Likelihood function

	$Y = 1$	$Y = 0$	Sum
$X = 1$	a	b	$a + b$
$X = 0$	c	d	$c + d$

- The likelihood function is

$$L(\boldsymbol{\beta}) = \prod_{i=1}^n \frac{e^{Y_i(\beta_0 + \beta_1 x_i)}}{1 + e^{\beta_0 + \beta_1 x_i}} = \frac{e^{\beta_0(a+c) + \beta_1 a}}{(1 + e^{\beta_0 + \beta_1})^{a+b} (1 + e^{\beta_0})^{c+d}}$$

The MLE estimator of β_i 's

- The log-likelihood function

$$\log L = (a + c)\beta_0 + a\beta_1 - (c + d)\log(1 + e^{\beta_0}) - (a + b)\log(1 + e^{\beta_0 + \beta_1})$$

- Setting derivatives to 0 yields

$$\frac{\partial \log L}{\partial \beta_0} = (a + c) - (c + d) \frac{e^{\beta_0}}{1 + e^{\beta_0}} - (a + b) \frac{e^{\beta_0 + \beta_1}}{1 + e^{\beta_0 + \beta_1}} = 0$$

$$\frac{\partial \log L}{\partial \beta_1} = a - (a + b) \frac{e^{\beta_0 + \beta_1}}{1 + e^{\beta_0 + \beta_1}} = 0$$

- Solving these two equations yields the MLE

$$\hat{\beta}_0 = \log\left(\frac{c}{d}\right) \quad \text{and} \quad \hat{\beta}_1 = \log\left(\frac{ad}{bc}\right).$$

Fisher Information Matrix

For $\theta = (\beta_0, \beta_1)^T$, the second-order derivatives yield Fisher Information Matrix:

$$\bullet \quad \frac{\partial^2 \log L(\theta)}{\partial \theta \partial \theta^T} = \begin{pmatrix} -\frac{(a+b)e^{\beta_0+\beta_1}}{(1+e^{\beta_0+\beta_1})^2} - \frac{(c+d)e^{\beta_0}}{(1+e^{\beta_0})^2} & -\frac{(a+b)e^{\beta_0+\beta_1}}{(1+e^{\beta_0+\beta_1})^2} \\ -\frac{(a+b)e^{\beta_0+\beta_1}}{(1+e^{\beta_0+\beta_1})^2} & -\frac{(a+b)e^{\beta_0+\beta_1}}{(1+e^{\beta_0+\beta_1})^2} \end{pmatrix}$$

$$\bullet \quad I(\hat{\theta}) = -\frac{\partial^2 \log L(\theta)}{\partial \theta \partial \theta^T} \Big|_{\hat{\theta}} = \begin{pmatrix} \frac{ab}{a+b} + \frac{cd}{c+d} & \frac{ab}{a+b} \\ \frac{ab}{a+b} & \frac{ab}{a+b} \end{pmatrix}$$

Variance of MLE

- By the general theory of the maximum likelihood estimator (MLE),

$$\text{Cov}(\hat{\boldsymbol{\theta}}) = I(\hat{\boldsymbol{\theta}})^{-1} = \begin{pmatrix} \frac{1}{c} + \frac{1}{d} & -\left(\frac{1}{c} + \frac{1}{d}\right) \\ -\left(\frac{1}{c} + \frac{1}{d}\right) & \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \end{pmatrix}$$

- Here we use the fact of the inverse of 2×2 matrix:

$$\begin{pmatrix} \alpha & \gamma \\ \beta & \delta \end{pmatrix}^{-1} = \frac{1}{\alpha\delta - \beta\gamma} \begin{pmatrix} \delta & -\gamma \\ -\beta & \alpha \end{pmatrix}$$

Are X and Y associated?

	$Y = 1$	$Y = 0$	Sum
$X = 1$	a	b	$a + b$
$X = 0$	c	d	$c + d$

- Thus $\hat{\beta}_1 = \log\left(\frac{ad}{bc}\right)$, $\widehat{\text{Var}}(\hat{\beta}_1) = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}$
- A 95% (approximate) confidence interval (CI) for β_1 is

$$\log\left(\frac{ad}{bc}\right) \pm 1.96 \sqrt{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}}$$

- **X is associated with Y if and only if 0 is outside this CI.**

Summary

The simplest logistic regression model

- Maximum Likelihood Estimators (MLE) for model parameters
- Asymptotic Variance of MLE
- Confidence interval of model parameters